



PxGrid Cloud 2.0: ISE – SCC integration

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Vision: Build a scalable, secure, and cost-effective cloud hosted platform on Security cloud

Business Impact: Enable seamless ISE integration with Security Cloud

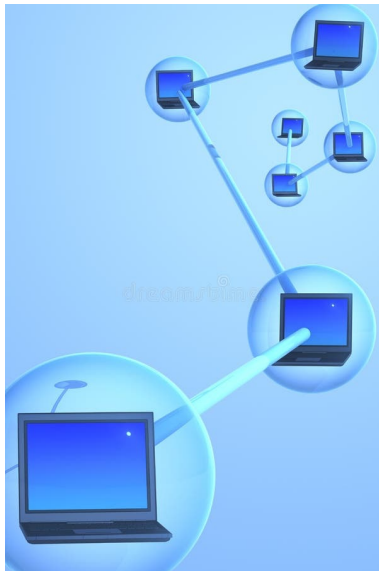
Key Outcomes:

- Onboard ISE device to security cloud
- Service Availability: 99.99%
- Response Time: <100ms
- Improved operational efficiency (Resource Utilization: >80%)
- Enhanced security and compliance

Expense Blueprint & Strategy

Network Data and Load

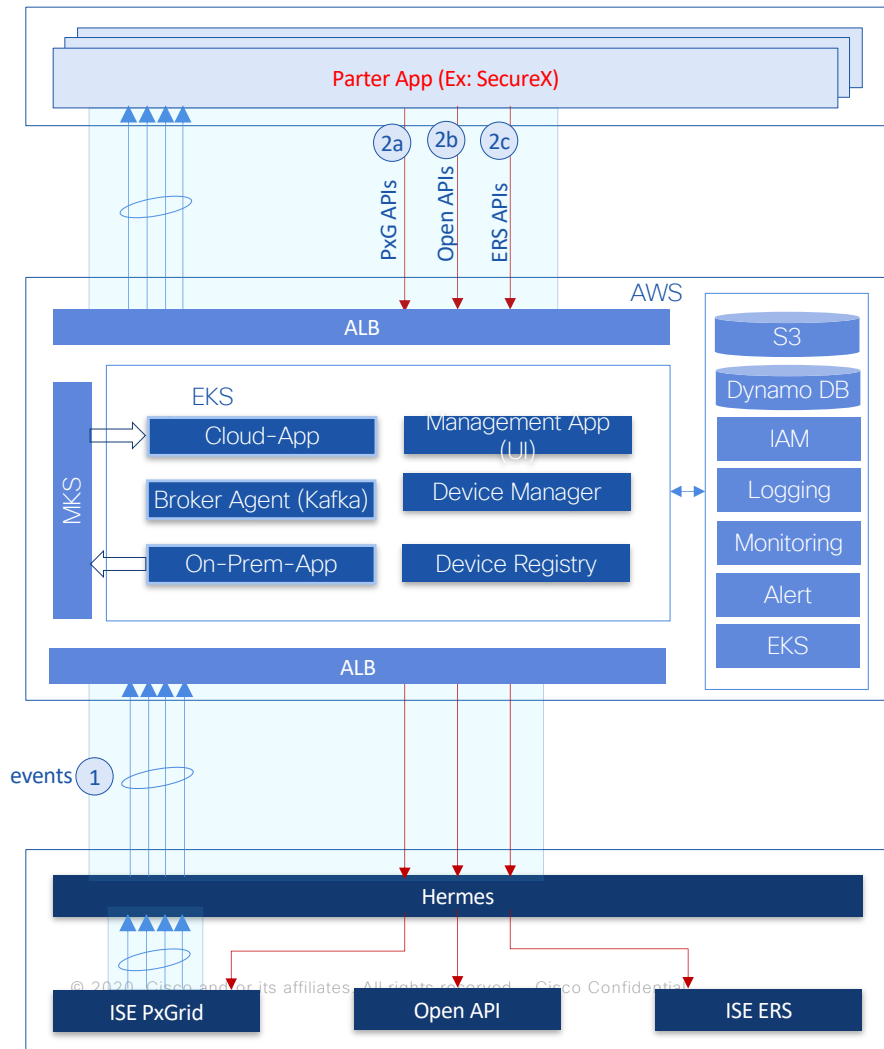
- Based on ISE Deployment
 - Small Deployment
 - Medium Deployment
- Based on Number of Consumers



Cloud infra and support

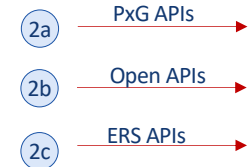
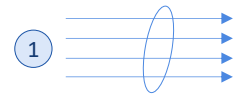


PxGrid Cloud – Data & Load Pattern & Cost Components



Load Types

Events - Message publishing and delivery



S.No	Scale Parameters	Count
1	Cloud Client per Tenant	3
2	Total Tenant	5000
3	Total End-points per Tenant	10,000

S.No	Cost Components
1	PxGrid Load: Message publishing and delivery (events)
2	PxGrid Load: API Load (PxG, OpenAPI, ERS)
3	PxGrid Load: Network-Throughput
4	EKS (Elastic Kubernetes service) - AWS Container Orchestration
5	Compute & Memory
6	Messaging: MKS (Managed Kafka Service)
7	Storage: DynamoDB, S3 (AWS Managed)
8	Observability: Monitor/Log/Alert (AWS Managed)
9	AWS Cloud Infra (VPC, NAT, R53, ALB)

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PxGrid Cloud – Data & Load Cost Forecast

• Region of deployment considered – Oregon (US-West2)

S.No	Cost Components	Cost/Year - USD Tenants (T) = 1 Consumer/T = 3 WITH LOAD	Cost/Year - USD Tenants (T) = 500 Consumer/T = 3 WITH LOAD	Cost/Year - USD Tenants (T) = 500 Consumer/T = 3 WITH OUT LOAD	Cost/Year – USD Tenants (T) = 1000 Consumer/T = 3 WITH LOAD	Cost/Year – USD Tenants (T) = 5000 Consumer/T = 3 WITH LOAD	Notes
1	PxGrid Load: - Message publishing and delivery (events) - API Load (PxG, OpenAPI, ERS)	\$111.02	\$46,942.55	\$65.64	\$91,485.34	\$447,826.84	API Requests/Day/Consumer = 189 Messages/Day/Consumer = 100010 \$9.25/M
2	Storage: DynamoDB, (AWS Managed)	\$60.48	\$2,029.68	\$263.16	\$3,565.68	\$18,695.28	Storage+write+read+Upfront = 60.48/Y
3	Storage: S3 (AWS Managed)	\$0.48	\$5,224.52	\$0.24	\$449.04	\$2,245.20	S3 Standard (Storage+GET+PUT) = \$0.04/M
4	AWS Container Orchestration – EKS (Elastic Kubernetes service)	\$876	\$4,380	\$876	\$8760	\$43,800	\$73/M
5	EKS EC2 – Compute & Memory	\$858.48	\$399,234	\$3193.92	\$798,468	\$3,992,340	\$71.54/M
6	Throughput	\$18.36	\$9,952.2	0	\$19,413.96	\$87,243.72	S3 Standard Internet = \$1.5/M3
7	Messaging: MSK (Managed Kafka Service)	\$1,222.25	\$49,556.04	\$1199.57	\$98,650.44	\$1,587,186.12	Broker-cluster + Storage + Data- Transfer = \$101.85/M
	TOTAL	\$3,147.07	512,318.99	\$5,598.53	\$1,020,792.46	\$6,179,337.16	

Assumptions

1	- Average API group requests per day = 189 - Average message events per day = 100,000 - End-points = 10,000; Messages per End-Point = 10	This estimated number may not be close to Production, we do not have the Production Load pattern yet or empirical/Test data for this offer.
3	Load Pattern/Day & Resource Elasticity	- This estimation involve average load that is continuously running (24H/D x 365.25). Bursts, Busy and Idle windows were not considered. - There could be idle window for 8h or 10h or 12h
4	Resource consumption (CPU, Mem, Disk)	Estimated based on previous project API load resource consumptions

PxGrid Cloud – other Infrastructure Cost Forecast

• *Region of deployment considered – Oregon (US-West2)*

S.No	Cost Components	Cost Type	Cost/Monthly - USD	Cost/Year - USD	Notes
1	AWS Cloud Infra:	Fixed Cost	\$320	\$3840	NAT Gateway: \$32, VPC Endpoints : \$100, Transit Gateway : \$73, Route 53 : \$15
2	Infrastructure Management	Fixed Cost	\$278	\$3336	Terraform State (S3) : \$3, Terraform State (DynamoDB) : \$2, CI/CD Pipeline : \$51, Multi-Region deployment : \$222
3	Security Tools	Fixed Cost	\$3269	\$39228	Security Hub: \$80, WAF : \$6, Shield Advanced : \$3000, GuardDuty : \$100, Secrets Manager : \$40, Certificate Management \$33, Access Management \$10
4	Monitoring	Variable Cost	\$1280	\$15360	CloudWatch Metrics \$200, CloudWatch Logs \$300, CloudWatch Alarms : \$50, Systems Manager : \$30, X-Ray : \$50, Managed Prometheus : \$200, Grafana : \$90, OpenSearch : \$200
5	API & Load Balancing	Variable Cost	\$300	\$3600	API Gateway : \$100 (Per million requests), Application Load Balancer : \$200 (Base + request costs)
6	Caching	Variable Cost	\$500	\$6000	ElastiCache (Redis): \$300 (2 nodes across regions), DynamoDB Accelerator : \$200 (Per node)
7	Container Management	Fixed Cost	\$130	\$1560	ECR: \$30 (Helm chart storage), ChartMuseum: \$100 (Self-hosted on EC2)
	Total		\$6,077	\$72,924	

Key Points

- **Fixed Cost (\$3,997/month):** These costs are incurred regardless of the number of tenants and include infrastructure, management, security, and container management tools.
- **Variable Cost (\$2,080/month per tenant):** These costs scale with the number of tenants, reflecting monitoring, API/load balancing, and caching expenses that increase with load.

PxGrid Cloud – overall Cost Forecast

• *Region of deployment considered – Oregon (US-West2)*

S.No	Cost Components	Cost/Year - USD Tenants (T) = 1 Consumer/T = 3 WITH LOAD	Cost/Year - USD Tenants (T) = 500 Consumer/T = 3 WITH LOAD	Cost/Year - USD Tenants (T) = 500 Consumer/T = 3 WITH OUT LOAD	Cost/Year – USD Tenants (T) = 1000 Consumer/T = 3 WITH LOAD	Cost/Year – USD Tenants (T) = 5000 Consumer/T = 3 WITH LOAD	Notes
1	PxGrid Cloud – Data & Load Cost Forecast	\$3,147.07	\$512,318.99	\$5,598.53	\$1,020,792.46	\$6,179,337.16	
2	PxGrid Cloud – other Infrastructure Cost Forecast	\$72,924	\$12,527,964	\$12,527,964	\$25,007,964	\$124,847,964	
	TOTAL	\$76,071.07	512,318.99	\$5,598.53	\$1,020,792.46	\$6,179,337.16	

Calculation Formula for “PxGrid Cloud – other Infrastructure Cost Forecast”

- Total Monthly Cost for T Tenants:
= Fixed Cost + (T × Variable Cost per Tenant)
- Total Annual Cost:
= Total Monthly Cost × 12

PxGrid Cloud – SRE Support

This section outlines the structure, skills, and responsibilities of the Site Reliability Engineering team. It details the technical and operational capabilities needed to effectively manage and evolve the platform, including on-call rotations, incident response, and continuous improvement processes.

Required skills:

- **Cloud Platform Expertise:** EKS and Kubernetes administration, RDS and database management, ElastiCache and Redis optimization, Route 53 and networking, IAM and security controls, CloudWatch and monitoring, AWS Cost Explorer and optimization.
- **Infrastructure as Code:** Terraform for infrastructure provisioning, AWS CloudFormation, Infrastructure version control, CI/CD pipeline management, GitOps workflows
- **Observability and Monitoring:** Prometheus and Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Distributed tracing (Jaeger/X-Ray), APM solutions, Custom metrics and dashboards.
- **Database Operations:** RDS operation and maintenance, Database performance tuning, Backup and recovery procedures, Replication management, Query optimization, Data lifecycle management.
- **Network Engineering:** VPC architecture and management, Direct Connect configuration, Load balancer optimization, Network security, Traffic routing and optimization, DNS management.
- **Security Operations:** AWS security best practices, Compliance frameworks, Security monitoring, Incident response, Vulnerability management, Access control and IAM

Core SRE Team (Minimum 5 members)

- **Lead (1):**
 - Architecture and strategy, Team leadership, Stakeholder management.
- **Senior SREs (2):**
 - Technical implementation, Mentorship, On-call leadership
- **SRE Engineers (2):**
 - Day-to-day operations, Implementation support, On-call rotation



Thank you



Backup

Average Load per day / Tenant (1 consumer)

API group	Description	Calls/Events per day	Entry size (bytes)	(Entries)	Message Size (bytes)	Total Bytes
ANC	config	10	100	10	1000	10000
ACI bindings	config	10	250	100	25000	250000
Egress matrix	config	10	350	10	3500	35000
IP/SGT mappings	config	10	200	100	20000	200000
SXP local bindings	config	10	200	100	20000	200000
SG	config	10	200	100	20000	200000
SG ACL	config	10	200	100	20000	200000
SG VN VLAN	config	10	1000	100	100000	1000000
Endpoint		10	1000	10000	10000000	100000000
pxGrid session get		10	1000	10000	10000000	100000000
pxGrid system health		24	350	12	4200	100800
pxGrid trustsec download		10	550	100	55000	550000
pxGrid trustsec config	config	10	200	100	20000	200000
pxGrid trustsec SXP		10	150	100	15000	150000
pxGrid profiler config	config	10	150	100	15000	150000
OpenAPI trustsec nbarapp	config	5	100	10	1000	5000
OpenAPI trustsec SG mapping/VN/VN mapping	config	20	100	100	10000	200000
Subtotal		189			20329700	203450800
Average request size (203450800/189) = 1.076Mb						
pxGrid session publish	websocket	100,000	1000	1	1000	100000000
pxGrid radius failure	websocket	10	5000	1	5000	50000
Subtotal		100010				100050000
Average Message size (100050000/100010) = 1Kb						

S.No	Ave Load per day	
1	API	
1.1	Ave Calls per day	189
1.2	Average request size (203450800/189)	1.076M
2	Message Events	
2.1	Ave Events per day	100010
2.2	Average Message size (100050000/100010)	1Kb
3	Total End-Points / Tenant	10,000

References

Tenants count	Cost Forecast details
1T with Load	https://calculator.aws/#/estimate?id=7b3fc49c701ad31a6c5b97754fcae1b130c21750
500T with Load	https://calculator.aws/#/estimate?id=3c067eeb2a8a31acf40cb71f82a04147c7c11782
500T with out Load	https://calculator.aws/#/estimate?id=2af83b6f30508beec5672e54f4bc9b55a59fa37d
1000T with Load	https://calculator.aws/#/estimate?id=6b1f65a71d38706772db09466358e617e5e7803f
5000T with Load	https://calculator.aws/#/estimate?id=5eaa26fcd97ca552cb5eb08912a6ebf8e90441d6
Cost estimation references: https://confluence-eng-rtp2.cisco.com/conf/display/CISE/pxGrid+Cloud+AWS+cost+estimates	

PxGrid Cloud – Cost Forecast based on 1 Device 1 consumer Model

S. No	Metric	Small Deployment (500K Endpoints)	Medium Deployment (1M Endpoints)	Notes
1	Total Endpoints	500,000	1,000,000	Direct input based on deployment size.
2	Estimated Concurrent Sessions	~10,000 (≈2% of endpoints)	~20,000 (≈2% of endpoints)	Assumes that about 2% of endpoints are active concurrently.
3	RADIUS/TACACS Transactions	~300 transactions per minute	~600 transactions per minute	For 500K endpoints, 300 transactions per minute represents roughly 0.06% of endpoints (300/500,000). This scales linearly with endpoints.
4	Log/Event Volume	~50,000 events per day	~100,000 events per day	Assumes 500K endpoints, ~50,000 events/day averages about 0.1 event per endpoint per day (~10% of endpoints generate an event daily); the medium deployment follows a similar ratio.
5	Daily Data Volume (Logs & Metrics)	~20–30 GB	~40–60 GB	Assume each endpoint generates ~40 KB of logs/metrics per day, then for a small deployment: 500,000 endpoints × 40 KB = 20,000,000 KB. Converting to GB: 20,000,000 KB ÷ 1,048,576 ≈ 19.1 GB, rounded to ~20 GB; overhead brings this to 20–30 GB. For 1M endpoints, the value doubles, yielding ~40 GB (scaled to 40–60 GB with overhead).
6	API Call Volume	~100–200 calls per minute	~200–400 calls per minute	Assumption: Roughly 0.02% of endpoints generate an API call each minute. For 500K endpoints, this yields ~100–200 calls per minute; for 1M endpoints, the call volume doubles to ~200–400 per minute.
7	Daily API Data Volume	~144,000–288,000 calls/day × 2 KB ≈ 0.288–0.576 GB	~288,000–576,000 calls/day × 2 KB ≈ 0.576–1.152 GB	Calculation: (calls per minute × 1440 minutes/day) × payload of 2 KB per call.
8	Pub/Sub Message Volume	~500–1,000 messages per minute	~1,000–2,000 messages per minute	Assumption: Approximately 0.1%–0.2% of endpoints generate a Pub/Sub message each minute. For 500K endpoints, that is ~500–1,000 messages per minute; doubling the endpoints to 1M results in ~1,000–2,000 messages per minute.
9	Daily Pub/Sub Data Volume	~720,000–1,440,000 messages/day × 1 KB ≈ 0.72–1.44 GB	~1,440,000–2,880,000 messages/day × 1 KB ≈ 1.44–2.88 GB	Calculation: (messages per minute × 1440 minutes/day) × 1 KB per message.
10	RADIUS/TACACS Data Volume	~432,000 transactions/day × 0.5 KB ≈ 0.216 GB	~864,000 transactions/day × 0.5 KB ≈ 0.432 GB	Calculation: 300 transactions/minute × 1440 minutes = 432,000 transactions/day at 0.5 KB each.

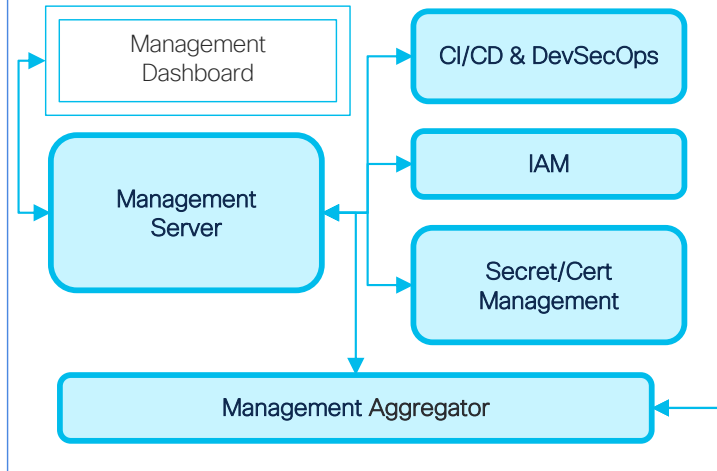
PxGrid Cloud – Cost Forecast based on 1 Device 1 consumer Model

S. No	Metric	Small Deployment (500K Endpoints)	Medium Deployment (1M Endpoints)	Notes
1	Total Daily Network Load	≈21.3–32.3 GB per day (Logs & Metrics + API Data + Pub/Sub Data + RADIUS/TACACS Data)	≈42.4–64.5 GB per day (Logs & Metrics + API Data + Pub/Sub Data + RADIUS/TACACS Data)	
2	Estimated Daily AWS Data Transfer Cost	≈\$1.92–\$2.91 (Calculated as Total Daily Network Load × \$0.09/GB; e.g., 21.3 GB × \$0.09 ≈ \$1.92; 32.3 GB × \$0.09 ≈ \$2.91)	≈\$3.82–\$5.81 (Calculated as Total Daily Network Load × \$0.09/GB; e.g., 42.4 GB × \$0.09 ≈ \$3.82; 64.5 GB × \$0.09 ≈ \$5.81)	Cost is calculated using an assumed rate of \$0.09 per GB for data transfer in the us-west-2 (Oregon) region for volumes under 10 TB/day.
3	Estimated Monthly AWS Data Transfer Cost (30 days)	≈\$57.6–\$87.3 (\$1.92–\$2.91 per day × 30 days)	≈\$114.6–\$174.3 (\$3.82–\$5.81 per day × 30 days)	Monthly cost is estimated by multiplying the daily cost by 30.

S. No	Parameter	Calculation/Assumption
1	Time Frame	All daily figures assume a 1440-minute day.
2	Payload Sizes	API Calls: 2 KB per call Pub/Sub Messages: 1 KB per message RADIUS/TACACS Transactions: 0.5 KB per transaction
3	Logs & Metrics Data Volume	Each endpoint is assumed to generate ~40 KB of log/metric data per day. For 500K endpoints: 500,000 × 40 KB = 20,000,000 KB Convert to GB: 20,000,000 KB ÷ 1,048,576 ≈ 19.1 GB, rounded up (with overhead) to ~20–30 GB. For 1M endpoints, the value approximately doubles (~40–60 GB per day).
4	API Call Volume	Assumes ~0.02% of endpoints generate an API call per minute. For 500K endpoints: 0.02% × 500,000 = 100 calls/min (lower bound) up to 200 calls/min (upper bound). For 1M endpoints, the value doubles to ~200–400 calls per minute.
5	Pub/Sub Message Volume	Assumes ~0.1%–0.2% of endpoints generate a Pub/Sub message per minute. For 500K endpoints: 0.1% yields 500 messages/min and 0.2% yields 1,000 messages/min. For 1M endpoints, these figures double to ~1,000–2,000 messages per minute.
6	RADIUS/TACACS Transactions	For 500K endpoints, estimated at ~300 transactions per minute (≈0.06% of endpoints). For 1M endpoints, this rate doubles to ~600 transactions per minute.
7	Cost Calculation	AWS data transfer cost is assumed at \$0.09 per GB (us-west-2, Oregon, for usage under 10 TB/day). Daily Cost = Total Daily Network Load (in GB) × \$0.09. Monthly Cost = Daily Cost × 30 days.
8	Scaling Assumptions	All metrics (API call volume, Pub/Sub messages, and RADIUS/TACACS transactions) scale linearly with the number of endpoints. For instance, if 500K endpoints yield ~300 RADIUS/TACACS transactions per minute (≈0.06% of endpoints), then 1M endpoints will yield ~600 transactions per minute.

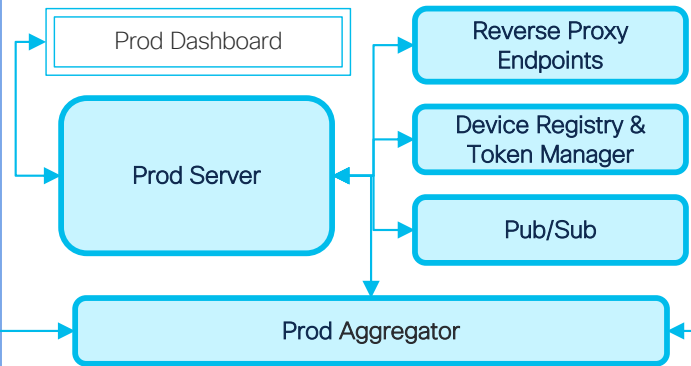
PxGrid Cloud 2.0 – Deployment Architecture

Management Environment

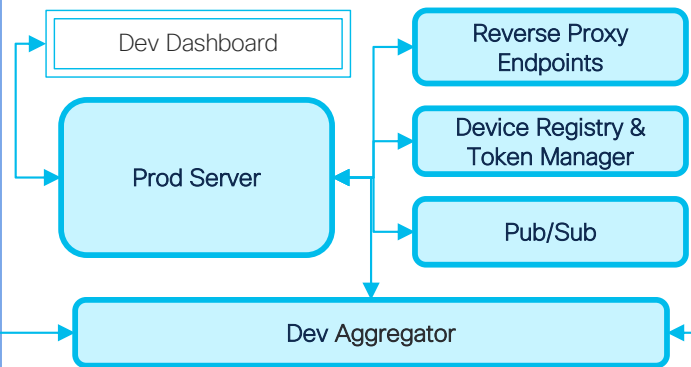


Provision & Manage

Prod Environment



Dev Environment



Common services

Platform Services (cross-cut) Environment

